

Lecture 18

- Complete Deutsch-Jozsa.
 - Bernstein - Vazirani
- worst case: first

- Probabilistic Algo

Choose input $\vec{x}$ randomly

if f is balanced, prob of

randomly getting $f(\vec{x})$ all the same in N queries

$P_M = 2 \times 2^{-M} \rightarrow$ probability of being fooled

$\nearrow$ 0 or 1 $\rightarrow$ getting 0 each time $> \frac{1}{2}$
 $\downarrow$
 1

this is query once then putting that back into the box.

Now no query without putting the input back in time

$$\tilde{P}_{\text{fail}} = 2 \cdot \left(\frac{\frac{N}{2}}{N} \right) \cdot \left(\frac{N_0 - 1}{N - 1} \right) \cdot \left(\dots \right)$$

$$\begin{aligned} N_0 &= N/2 \\ N_1 &= N/2 \\ &= \frac{(N_0 - 1)}{(N - 1)} \cdot \frac{(N_0 - 2)}{(N - 2)} \cdot \dots \cdot \frac{(N_0 - (M - 1))}{(N - (M - 1))} = 2 \prod_{k=0}^{M-1} \frac{N/2 - k}{N - k} \approx 2 \left(\frac{1}{2} \right)^M \\ &= 2^{1-M} \end{aligned}$$

$$M \ll N \Rightarrow \frac{N_0 - M}{N - M} = \frac{\frac{1}{2} - \frac{M}{N}}{1 - \frac{M}{N}} \approx \left(\frac{1}{2} - \frac{M}{N} \right) \left(1 + \frac{M}{N} \right)$$

taylor approx.

$$= \frac{1}{2} + \frac{M}{2N} - \frac{M}{N} - \frac{M^2}{N^2}$$

$$= \frac{1}{2} - \frac{M}{2N} - \cancel{\frac{M^2}{N^2}}$$

$$= \frac{1}{2} \left(1 - \frac{M}{N} \right)$$

$$2 \cdot \prod_{k=0}^{M-1} \frac{N/2 - k}{N - k} \approx 2^{1-M} \prod_{k=0}^{M-1} \left(1 - \frac{k}{N} \right)$$

$$\prod_{k=0}^{M-1} \left(1 - \frac{k}{N} \right) \approx \exp \left(- \frac{1}{N} \sum_{k=0}^{M-1} k \right) = \exp \left(- \frac{M(M-1)}{2N} \right)$$

$$\hookrightarrow \boxed{2^{1-M} \exp \left(- \frac{M(M-1)}{2N} \right)}$$

Deutsch-Jozsa $\rightarrow$ in the query complexity class EQP

(exact quantum polynomial)



exponentially better than the exact classical
algorithm.

$\downarrow$
exponential

classically complexity BPP

Bounded error probability polynomial time.

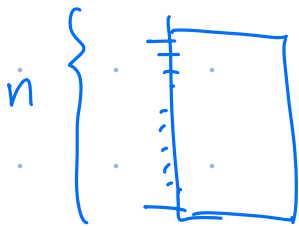
- Bernstein-Vazirani: bounded error Quantum polynomial time BQP

BU doesn't show exp. speed up?

Follows Rietel & Polak see 7.5.2

$$f(\vec{x}) = \vec{u} \cdot \vec{x}$$

fixed, hidden bit string



Classical Query Complexity
 $\rightarrow$ each query yields 1 bit,
 hence $n_{\text{query}} \geq n$

Proof that $M_{\text{optimal}} = n$.

$$\vec{x}_0 = (000 \dots 001) \quad f(\vec{x}_0) = \vec{x}_0 \cdot \vec{u} = u_0$$

$$\vec{x}_1 = (000 \dots 010) \quad f(\vec{x}_1) = \vec{x}_1 \cdot \vec{u} = u_1$$

$$\vec{x}_{n-1} = (100 \dots 000) \quad f(\vec{x}_{n-1}) = \vec{x}_{n-1} \cdot \vec{u} = u_{n-1}$$

$\rightarrow$ polynomial complexity.

$$|\psi_2\rangle = 1 \rightarrow \otimes \frac{1}{2^n} \sum_{\vec{x}, \vec{z}} (-1)^{(\vec{u} \oplus \vec{z}) \cdot \vec{x}} |\vec{z}\rangle$$

conservation of probability argument

$$\text{if } \vec{z} = \vec{u} \Rightarrow |\psi_1\rangle = 1 \rightarrow \otimes |\vec{u}\rangle$$

with probability 1

so all the others must be 0

Balance func. then.

$$q(\vec{x}) = \vec{y} \cdot \vec{x} \quad \text{balance of } \vec{y} \neq 0$$

David Mermin: Control & Target

what if $\vec{x} = 0$
 doesn't the machine
 gets that
 off as well